Gene Essentiality from the Cell Graph

The backbone against flux balance analysis and Flux Cone Learning,

and what routing the prediction through the metabolic module does

Michael Volk

September 16, 2026

Abstract

Gene essentiality is the one endpoint on which a flux model has a published, non-trivial score: on the released Merzbacher 2025 yeast split, flux balance analysis reads an AUROC of 0.69 over all 1,121 labeled genes and Flux Cone Learning 0.73 on the 223 held-out genes. This document scores the cell graph transformer on the same held-out genes, puts FBA on every medium the ontology defines, and asks two questions of the metabolic module: whether a prediction read off its growth column ranks essential genes better than the backbone alone, and where the gradient goes when the module is one of two paths rather than the only one. Every number here is at a validation-selected epoch on a pinned test set, with a gene bootstrap and the across-seed range, and every table is generated from the result files.

Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

Contents

1 The endpoint, its labels and its splits ✗ . . . **2**
1.1 Vocabulary ✗ . . . 2
1.2 Where the labels come from ✗ . . . 2
1.3 The protocol inherited from 027, and what it got wrong ✗ . . . 2
2 The comparators, computed rather than quoted ✗ . . . **2**
2.1 Flux balance analysis on four media ✗ . . . 2
2.2 Flux Cone Learning on the same genes ✗ . . . 3
3 The backbone: splits, labels, weight decay and the shape of the curve ✗ . . . **3**
4 Routing the prediction through the metabolic module ✗ . . . **3**
5 Open questions ✗ . . . **3**

1 The endpoint, its labels and its splits ✗

1.1 Vocabulary ✗

AUROC Area under the receiver operating characteristic, computed as the Mann-Whitney rank statistic with mid-ranks on ties: the probability that a randomly drawn essential gene scores above a randomly drawn nonessential one.

backbone The cell graph transformer over the nine-graph perturbed gene graph with `prot_T5_all` node features, read at the pooled perturbed tokens. The *pooled* arm is this backbone with a scalar head and no metabolic module.

gene bootstrap Resampling the scored genes with replacement (2,000 draws) and reporting the 2.5 and 97.5 percentiles of the statistic.

held-out genes The 223 genes Merzbacher 2025 held out of its own training (`test = 1` in the released split), of which 195 resolve as single-deletion genotypes in the build. Every model number in this document is on those 195 unless the gene set is named.

metabolic module The enzyme-constrained thermodynamic flux layer over yeast-GEM 9.0.2 that maps the perturbed representation to a flux vector; here it is read at the objective column `r_2111` (growth).

released labels The 1,121 essentiality calls in the released split. The column named `essential` is a viability flag (1.0 marks a nonessential gene); it is inverted once and the four class counts are asserted against the paper's Methods.

selected epoch The argmax of a centered five-epoch moving average of the validation AUROC. The held-out genes are scored once, at that epoch.

1.2 Where the labels come from ✗

The build is a union of the SGD essential-gene list (1,140 genotypes, all essential) and the single-mutant fitness screens (Costanzo 2016, Kuzmin 2018; 4,540 genotypes), 5,656 single-deletion genotypes in all. No fitness value enters any objective: the head is one logit scored by binary cross entropy against a 0/1 label, and every genotype whose gene carries no label arrives masked. Two label sources are used and named throughout:

Table 1: The split files, one per label source: genes per split with the essential count in parentheses. The test block is the released held-out genes under either source; validation is a class-stratified fifth of the remaining labeled genes at seed 0. Under the genome source the released label wins on every disagreement (12 genes).

split file	labels	train (ess)	val (ess)	test (ess)
`ess_fcl_val0.2_s0`	released (1,121)	629 (100)	158 (25)	195 (31)
`ess_genome_val0.2_s0`	released + SGD essential + viable fitness (5,795)	4,368 (879)	1,093 (220)	195 (31)

1.3 The protocol inherited from 027, and what it got wrong ✗

The wave-9 protocol pinned the held-out genes and let the data module draw train and validation at 80/10/10 over the remaining 5,461 records. Of those, 926 carry a label, so the validation set held 75 labeled genes with about 12 essential, on which a single AUROC read has a Hanley-McNeil standard error near 0.07; 10 percent of the labeled genes sat in a random test block nothing reports; and 4,535 unlabeled records were forwarded every epoch under a False mask. A split file replaces the draw: it names every record's split, is class-stratified, carries the labels it was stratified on, and the trainer drops every record it does not name.

2 The comparators, computed rather than quoted ✗

2.1 Flux balance analysis on four media ✗

The published FBA comparator is cobrapy `single_gene_deletion` with glucose as the carbon source, scored on all 1,121 labeled genes. The phenotype it is compared against was called on rich medium, and the model here is scored on the 195 held-out genes, so the comparison as published is neither on the same medium nor on the same genes. Table 2 recomputes it on yeast-GEM 9.0.2 under the exchange bounds the model ships with and under the three media the ontology defines, with the essentiality score being the negative knockout-to-wild-type growth ratio, rounded to 10^{-3} so

that the roughly 900 deletions that grow at 0.99 to 1.00 of the wild type rank as the tie they are (without the rounding two identical runs differed by 0.03 AUROC on the held-out genes).

Table 2: FBA single-gene deletion on yeast-GEM 9.0.2 (cobrapy 0.30.0), scored as the negative knockout-to-wild-type growth ratio rounded to 10^{-3}. AUROC on all 1,121 labeled genes, on the 223 held-out genes, and on the 195 held-out genes the model scores with a 2,000-draw gene bootstrap interval. 1121 of the labeled genes are in the GEM; the rest keep wild-type growth.

medium	WT growth (1/h)	lethal	all 1,121	held-out 223	our 195 [95% CI]
shipped bounds (glucose minimal, aerobic)	0.086	186	0.707	0.633	0.654 [0.56, 0.75]
SM, synthetic minimal	0.314	160	0.707	0.642	0.666 [0.57, 0.76]
SC, synthetic complete	0.543	91	0.719	0.653	0.666 [0.57, 0.76]
YPD approximation (no peptone)	0.535	104	0.715	0.653	0.664 [0.57, 0.76]

The medium moves the number of lethal deletions by a factor of two and the AUROC by about 0.01. The published 0.69 on all labeled genes is reproduced to within 0.02.

2.2 Flux Cone Learning on the same genes ✗

Recomputed from the authors' released per-gene predictions: 0.7308 on the 223 held-out genes (published 0.73) and 0.742 on the 195 genes the model scores. On those same 195 genes the released FBA predictions score 0.640, so the published pair overstates FBA by about 0.06 relative to a head-to-head on one gene set.

3 The backbone: splits, labels, weight decay and the shape of the curve ✗

Filled from the wave-12 readout when the round returns.

4 Routing the prediction through the metabolic module ✗

Filled from the wave-13 readout when the round returns.

5 Open questions ✗

Filled as the rounds return.